

**Experiment 15: Iris Flower Classification using Naïve Bayes****Aim**

To implement **Iris Flower Classification using the Naïve Bayes classifier** in Python.

**Algorithm**

**Step 1:** Import the required libraries.

**Step 2:** Load the Iris dataset.

**Step 3:** Separate the input features and target classes.

**Step 4:** Split the dataset into training and testing sets.

**Step 5:** Create a Gaussian Naïve Bayes classifier.

**Step 6:** Train the classifier using the training data.

**Step 7:** Predict the flower classes for the test data.

**Step 8:** Calculate and display the accuracy.

**Program**

```
# Experiment 15 - Iris Flower Classification using Naive

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

### Output

```
... Predictions: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]  
Accuracy: 1.0
```

### Result

Thus, the **Iris Flower Classification using the Naïve Bayes classifier** was successfully implemented, and the flower classes were predicted for the test data.